

Bank Loan Case Study

Project Description

First thing that comes to my mind when I think about bank loan is “huge process” as everyone would so far know about bank loan process, if not, Banks typically lend money to individuals or businesses based on their financial health, creditworthiness, and other relevant factors for the purposes like buying homes, starting or expanding businesses, or consolidating debt etc. However, lending money comes with risks, particularly the risk of default, where the borrower fails to repay the loan due to financial distress, job loss, high debt levels, or poor financial management, market risk-Changes in the broader economy, such as recessions, inflation, or currency fluctuations, can affect borrower’s ability to repay loans as well poor customer service can negatively affect the bank's reputation and can influence other people too & The use of fake documentation, identity theft, or employees intentionally approving fraudulent loans. This is the reason why it’s a huge process, because it has various criteria to invest and us being fit in to all those criteria is a one such thing so, this case study aims to go through various factors that influence loan approval, denial, and default risk, it Examines different factors like credit score, income, loan amount and also understand how banks lighten risks through better decision-making and data analysis

My task is to use Exploratory Data Analysis (EDA) to analyze patterns in the data and ensure that capable applicants are not rejected.

When a customer applies for a loan, your company faces two risks:

1. If the applicant can repay the loan but is not approved, the company loses business.
2. If the applicant cannot repay the loan and is approved, the company faces a financial loss.

The dataset I'll be working with contains information about loan applications. It includes two types of scenarios:

1. Customers with payment difficulties: These are customers who had a late payment of more than X days on at least one of the first Y installments of the loan.
2. All other cases: These are cases where the payment was made on time.

When a customer applies for a loan, there are four possible outcomes:

1. Approved: The company has approved the loan application.
2. Cancelled: The customer cancelled the application during the approval process.
3. Refused: The company rejected the loan.
4. Unused Offer: The loan was approved but the customer did not use it.

My goal in this project is to use EDA to understand how customer attributes and loan attributes influence the likelihood of default.

Approach

The primary goal of this case study is to identify patterns that indicate if a customer will have difficulty paying their installments. This information can be used to make decisions such as denying the loan, reducing the amount of loan, or lending at a higher interest rate to risky applicants. The company wants to understand the key factors behind loan default so it can make better decisions about loan approval.

As the goal is very clear, my approach would include:

- Firstly, data cleaning (null values, blank cells, handling missing values, handling outliers, data imbalance....)
- Dropping of columns containing null proportion having more than 45%
 - Sort the data to do analysis
- I performed Descriptive Statistics on numerical variables (mean, median, standard deviation, etc.)
 - I majorly used Data Visualization (Usage of graphs and charts (histograms, bar charts, scatter plots) to visualize relationships between factors such as income, credit score, and loan approval etc.
- I did the Correlation Analysis to understand relationships between variables

By following this approach, you can precisely analyze the bank's loan processes, identify key risk factors, and develop data-driven strategies to improve loan approval rates, predict defaults more accurately, and enhance overall financial stability. The final case study report should provide the bank with valuable insights.

Tech-Stack Used

I've used the following

- Microsoft Excel for the (Data cleaning, visualizing, interpreting & analysis)
- Word to provide description about project showcasing insights & results
- Canva to do the ppt to go through project at ease

INSIGHTS

Identify Missing Data and Dealing with it Appropriately

First step is to clean to data this time I did the EDA (Exploratory Data Analysis) for that first is to remove blank cells, null values and handle missing values This time I choose other approach which is to count the blank cells and their percentage, if the percentage is more than 45% I dropped all the columns cause its waste of time to analyse the data anyway so I removed them if not in use

To count blank cells: =COUNTBLANK(range)

To count blank percentage = (Number of missing values / Total number of values) *100

With that count I've highlighted columns which are less than 45%with green colour

And more than 45% I've highlighted with orange colour and dropped them to make data less complicate and more useful

Also changed columns having days in negative
ABSOLUTE Value = ABS(number)

And made it visually also created a column chart to visualize the proportion of missing values for each variable to understand better and easily

And for handling missing values for null proportion having less than 45%, for word categories I used mode and for numeric variables I used mean &median

RESULT

EDA, NULL PROPORTION MISSING VALUES & BLANK CELLS.xlsx

INSIGHTS

Identify Outliers in the Dataset

Once I got the cleaned data the task was to detect outliers, outliers are data points which deviate from the majority of the data. They can be caused by errors, anomalies etc...

So with the help of conditional formatting & excel functions like quartile, IQR to identify potential outliers from the dataset I possibly detected 8 columns which had outliers with the columns then calculated both lower(Q1) and upper quartile(Q3) and also IQR (difference between upper & lower quartile) and also found the upper limit and lower limit of quartiles

To calculate Q1&Q3: =QUARTILE (range, quart)

To calculate IQR: Q3-Q1

For the upper and lower limit, a commonly used rule says that a data point is an outlier if it is more than $1.5 * \text{IQR}$ above the third quartile or below the first quartile.

low outliers are below $Q1 - 1.5 * \text{IQR}$

high outliers are above $Q3 + 1.5 * \text{IQR}$

And then for all the detected columns, I calculated descriptive statistics and also to visually understand used box plots or scatter

plots to visualize the distribution of numerical variables and to highlight the outliers

RESULT

FINAL DATA & OUTLIER DETECTION.xlsx

INSIGHTS

Analyze Data Imbalance

I selected target column where I suspected data could be imbalanced so to know the ratio of 0&1

with the help of countif I calculated count of target, where data was either 0 or 1

0 represents customers paying on time=NON-DEFAULTERS

1 represents customers having issues regarding payments=DEFAULTERS

To know the ratio =number1/number2

And the result was pretty obvious data is imbalanced which is of having 0 at 92% &1 at 8%

And to client contract type either is it cash loans (a loan which is received by the borrower in cash) or revolving loans(is a type of loan that allows a borrower to withdraw funds up to a pre-approved limit, repay them, and then withdraw them again Ex: credit cards) and with countif formula I calculated clients having cash loans and clients having revolving loans

And result was as clients having cash loans with 91%& clients having revolving loans with 9%

Represented found data with variety of charts with the help of graphical tools

RESULT

[ANALYSE DATA IMBALANCE.xlsx](#)

INSIGHTS

Perform Univariate, Segmented Univariate, and Bivariate Analysis

univariate analysis to understand the distribution of individual variables, segmented univariate analysis to compare variable distributions for different scenarios, and bivariate analysis to explore relationships between variables and the target variable using Excel functions and features

firstly I've done the segmented univariate which mostly included amount cause we can't calculate each and every amount so it's better to give a range accordingly by filtering the column we can know the highest and lowest range by that we can give our range and calculated total clients by countif and also calculated non defaulters and defaulters with the help of target column, having 0 & 1 which defines 0 is people are paying on time and 1 is people who are having issues regarding payment

the columns I've done segmented univariate are as follows

- AMT_INCOME_TOTAL
- AMT_CREDIT
- AMT_ANNUITY
- AMT_GOODS_PRICE

And Univariate data is a term to describe data that consists of observations on only one characteristic or attribute so the columns I've found are

- CODE_GENDER
- NAME_FAMILY_STATUS
- NAME_EDUCATION_TYPE
- OCCUPATION_TYPE

And used all types of graphical tools in excel to represent creatively

Bivariate analysis: is a statistical method used to investigate the relationship between two variables.

I've chosen the columns which are interlinked which are

- AMT_CREDIT, AMT_ANNUITY & AMT_INCOME_TOTAL

Here client total income, annual income definitely plays a major role in amount of credit he can access it shows a relationship 2 different variables

Used clustered bar & line chart to represent creatively

- Client_Age, AMT_CREDIT & TARGET

Here client age and the amount of credit person gets according to age, target is nothing but having 0 & 1 which defines 0 is people are paying on time and 1 is people who are having issues regarding payment

Calculated average of amt_credit with average excel function

Countif for total applicants, non-defaulters & defaulters and also represented data using variety of charts.

RESULT

UNIVARIATE & BIVARIATE.xlsx

INSIGHTS

Identify Top Correlations for Different Scenarios

Scenario 1:

Finding out correlation with target column first with all the non-defaulters (0), 0 represents people paying on time, then filtering out target column with 0 only then selecting some of the columns to find out correlation between them, first after selecting columns I've written all the columns horizontally and vertically to find the correlation and with the excel `CORREL(Array1,Array2)` found out correlation between columns

Here 1 represent perfect positive correlation (highlighted with yellow color)

And the next top correlations I've written them down numerically highlighted with green color

Scenario 2:

Finding out correlation with target column again but with all the Defaulters (1), 1 represents people having payment issues, then filtering out target column with 1 only then selecting some

of the columns to find out correlation between them, first after selecting columns I've written all the columns horizontally and vertically to find the correlation and with the excel CORREL (Array1, Array2) found out correlation between columns

Here 1 represent perfect positive correlation (highlighted with yellow colour)

And the next top correlations I've written them down numerically highlighted with green colour

RESULT

[CORRELATION.xlsx](#)

